In this section, we'll focus on optimizing the route of a delivery truck, assuming its load (and in turn its destinations) are already fixed. Once we have improved this phase, we still have another, higher-level optimization in front of us: assigning the deliveries to the trucks to minimize the distance (or travel time, or, more likely, cost) across all trucks and all deliveries. But we need to take one step at the time, so in this chapter we'll focus on the following problem: given a list of cities, each connected to each other by roads, find the optimal tour, aka a sequence of the cities such that we move from each city to the next one, at some cost (for instance the distance between the two cities) and eventually return to the first city, while keeping the total cost minimal.

Figure 17.8 illustrates this situation: you can see ten among real cities and DC-Universe cities, the connections between each pair of cities, and – highlighted – the shortest tour touching each city exactly once.

Now, we are considering a single delivery per city just to make the example clearer, with all steps being at the same scale: nothing changes, however, if instead we have multiple deliveries in each city: since the intra-city distances will be smaller than the ones between different cities, deliveries in the same city will be naturally clustered together; it is also likely possible for the problem to be optimized at different levels in two steps: first find the best order in which cities should be visited, and then within each city compute the best route, using the same algorithm[10]. But this are just low-level details.